Context Free Grammars

Let the terminals be a and b, the only nonterminal be S, and the productions be

Prod1 S → aS

Prod2 S → bS

Prod3 S → Λ

The word *ab* can be generated by the derivation

S => aS => abS => abΛ = ab

The word baab can be generated by

S => bS => baS => baaS => baabS => baabΛ = baab

Clearly, the language generated by the above CFG is

(a + b)*.

Let the terminals be a and b, the only nonterminal be S, and the productions be

Prod1 S → aS

Prod2 S → bS

Prod3 S → a

Prod4 S → b

The language generated by the above CFG is

(a + b)+.

Example

Let the terminals be a and b, the the nonterminal be S and X, and the productions be

Prod 1 $S \rightarrow XaaX$

Prod 2 $X \rightarrow aX$

Prod 3 $X \rightarrow bX$

Prod 4 $X \rightarrow \Lambda$

We already know from the previous example that the last three productions will generate any possible strings of a's and b's from the nonterminal X. Hence, the words generated from S have the form

anything aa anything

Hence, the language produced by this CFG is

(a + b)*aa(a + b)*

which is the language of all words with a double a in them somewhere.

For example, the word *baabb* can be generated by

S → XaaX → bXaaX → baaX → baaX

→ baabX → baabbX → baabbΛ = baabb

Example

Consider the CFG:

$S \rightarrow aSb$

$S \rightarrow \Lambda$

It is easy to verify that the language generated by this CFG is the **non-regular** language $\{a^n b^n\}$.

For example, the word a^4b^4 is derived by

$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aaaSbbb$

$\Rightarrow aaaaSbbbb \Rightarrow aaaa\Lambda bbbb = aaaabbbb$

Disjunction Symbol |

Let us introduce the symbol | to mean disjunction (or).
We use this symbol to combine all the productions that have the same left side.
For example, the CFG

Prod 1 $S \rightarrow XaaX$

Prod 2 $X \rightarrow aX$

Prod 3 $X \rightarrow bX$

Prod 4 $X \rightarrow \Lambda$

can be written more compactly as

Prod 1 $S \rightarrow XaaX$

Prod 2 $X \rightarrow aX|bX|\Lambda$